Possible Problems for Quiz 6 to be taken 2/21/19. Of these problems I will choose 3 to be on the quiz.

Problem 1

Let $A=\begin{bmatrix}1 & 1 & -1\\ 0 & 0 & 2\\ -1 & 2 & 1\end{bmatrix}$ find $\det(A)$. Is A invertible?

Problem 2

What is the Rank(A) for $A=\begin{bmatrix}1 & 1 & -1\\ 0 & 1 & 2\\ 0 & 0 & 1\end{bmatrix}$? Use this knowledge to determine all the vectors in the nullspace of A. Justify your answers.

Problem 3

Let $[\vec{x}]_{B_1}=\begin{bmatrix}1\\2\\3\end{bmatrix}$ the basis $B_1=\left\{\begin{bmatrix}1\\0\\1\end{bmatrix},\begin{bmatrix}-1\\2\\1\end{bmatrix},\begin{bmatrix}1\\1\\0\end{bmatrix}\right\}$ What is $\vec{x}$?

Problem 4

Suppose A is a triangular matrix of size $n\times n$ where $a_{ii}=2^i$. What is the Determinant of A?

Problem 5

Write down a 4×4 matrix which has no zero entries yet has determinant 0. Is your matrix invertable?

Problem 6

If A=$\begin{bmatrix}-1 & 1 & 2 & 4 & 2\\ -5 & 3 & 1 & 1 & 0\\ 4 & 0 & 0 & 1 & -1\\ 3 & -4 & 1 & 5 & 0\\ 0 & 2 & 2 & 5 & -1\end{bmatrix}$ and $\det(A)=34$ find the following:

$$\det(\mathrm{B})=\det\left(\begin{bmatrix}-1 & -5 & 4 & 3 & 0\\ 1 & 3 & 0 & -4 & 2\\ 2 & 1 & 0 & 1 & 2\\ 4 & 1 & 1 & 5 & 5\\ 2 & 0 & -1 & 0 & -1\end{bmatrix}\right)$$

$$\det(\mathrm{C})=\det\left(\begin{bmatrix}1/17 & -7/17 & -2/17 & -3/17 & 4/17\\ 4/17 & 23/17 & 26/17 & 5/17 & -18/17\\ -12/17 & -137/17 & -275/34 & -49/17 & 227/34\\ 5/17 & 50/17 & 99/34 & 19/17 & -79/34\\ 9/17 & 22/17 & 49/34 & 7/17 & -47/34\end{bmatrix}\right)$$

For the second on use the additional information that $AC=I$.

Problem 7

Let $Rank(A)=100$ and $Nullity(A)=50$ how many columns does A have?